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Saveetha Institute of Medical And Technical Sciences

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ITA0513 – COMPUTER VISION

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ASSIGNMENT – 1

Overview of Computer Vision

Course Code	Course Name	Assessment	CO	BTL	Max. Marks
ITA0513	Computer Vision	Assignment – 1	CO1	BTL3	50

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analysing Computer Vision systems.

Instructions: Answer all five questions (10 marks each). Show every computational step, state formulas before substituting values, and justify each decision using your computed results.

Mark Allocation

Q. No.	Topic	(a)	(b)	(c)	(d)	Total
1	Sampling, quantization and data rate	3	3	2	2	10
2	Pinhole camera projection	3	3	2	2	10
3	Illumination–reflectance model	3	3	2	2	10
4	Pixel adjacency and connectivity	3	3	2	2	10
5	Levels of vision and applications	3	3	2	2	10
	Total	15	15	10	10	50

Question 1 (CO1, BTL3, 10 Marks) – Sampling and Quantization for Blister-Pack Inspection

A pharmaceutical plant inspects aluminium blister packs for pinhole cracks using a monochrome machine-vision camera. The sensor has 2448 × 2048 pixels, a native bit depth of 12 bits and a maximum frame rate of 60 fps. The optics image a field of view of 245 mm × 205 mm onto the full sensor. The smallest crack that must be detected is 0.25 mm wide and produces an intensity contrast of only 1.5 % of full scale. The production line delivers 600 blister packs per minute, and one frame is captured per pack. Two interfaces are available: GigE Vision with an effective throughput of 115 MB/s and USB3 Vision with an effective throughput of 380 MB/s (1 MB = 10^6 bytes).

(a) Compute the spatial sampling interval (mm/pixel) in both directions and determine whether the 0.25 mm crack satisfies the Nyquist requirement of at least two pixels across the defect. [3 Marks]
(b) Compute the raw frame size for 12-bit packed data and the data rate at 60 fps. Determine whether either interface can sustain full-rate 12-bit capture. [3 Marks]
(c) Compute the number of grey levels spanned by the 1.5 % crack contrast and the theoretical signal-to-quantization-noise ratio (SQNR ≈ 6.02b + 1.76 dB) for 8-bit and 12-bit quantization. [2 Marks]
(d) The line needs only 600/60 = 10 fps. Compute the maximum frame rate of GigE for 12-bit and 8-bit frames, and justify the most economical camera configuration that meets both the throughput and the crack-contrast requirements. [2 Marks]

ANSWER:

Digital Image Fundamentals

(a) Spatial Sampling Interval and Nyquist Requirement

Given:

- Sensor resolution = 2448 × 2048 pixels
- Field of View (FOV) = 245 mm × 205 mm
- Crack width = 0.25 mm

Formula:

\ [\text {Sampling interval} =\frac {\text {FOV}} {\text {Number of pixels}} \]

Horizontal direction:

\ [=\frac {245} {2448} =0.10008\text {mm/pixel} \]

Vertical direction:

\ [=\frac {205} {2048} =0.10010\text { mm/pixel} \]

Number of pixels across the crack:

\[=\frac{0.25} {0.10008}=2.498\text{ pixels} \]

Result: The crack covers approximately 2.5 pixels. Since it is greater than 2 pixels, it satisfies the Nyquist requirement.

(b) Raw Frame Size and Data Rate [3 Marks]

Given:

- Resolution = 2448 × 2048
- Bit depth = 12 bits
- Frame rate = 60 fps

Formula:

Raw Frame Size and Data Rate [3 Marks]

Given:

- Resolution = 2448 × 2048

- Bit depth = 12 bits
- Frame rate = 60 fps

Formula:

$$\text{Frame size}=\frac{W\times H\times b}{8}$$

$$=\frac{2448\times2048\times12}{8}$$

$$=7,520,256\text{ bytes}$$

$$=7.52\text{ MB/frame}$$

Data rate at 60 fps:

$$=7.52\times60$$

$$=451.22\text{ MB/s}$$

Interface comparison:

- GigE Vision = 115 MB/s
- USB3 Vision = 380 MB/s

Since 451.22 MB/s is greater than both interface throughputs, neither interface can support full-rate 12-bit capture at 60 fps.

(c) Grey Levels and SQNR [2 Marks]

Given:

- Crack contrast = 1.5% of full scale.

For 8-bit quantization:

Number of grey levels:

$$=0.015\times255=3.825$$

$$\approx 4\text{ grey levels}$$

SQNR formula:

$$SQNR=6.02b+1.76$$

$$=6.02(8)+1.76$$

$$=\boxed{49.92\text{ dB}}$$

For 12-bit quantization:

Number of grey levels:

$$=0.015\times4095=61.425$$

$$\approx61\text{ grey levels}$$

SQNR:

$$=6.02(12)+1.76$$

$$=\boxed{73.99\text{ dB}}$$

Result: 12-bit quantization provides more grey levels and better intensity resolution for detecting low-contrast cracks.

(d) Maximum Frame Rate and Economical Configuration [2 Marks]

Formula:

$$FPS=\frac{\text{Interface throughput}}{\text{Frame size}}$$

1. GigE with 12-bit frames:

$$=\frac{115}{7.520256}$$

$$=\boxed{15.29\text{ fps}}$$

2. GigE with 8-bit frames:

8-bit frame size:

$$=\frac{2448\times2048\times8}{8}$$

$$=5,013,504\text{ bytes}=5.01\text{ MB}$$

Maximum frame rate:

$$=\frac{115}{5.013504}$$

$$=\boxed{22.94\text{ fps}}$$

Economical camera configuration:

- Required frame rate = 10 fps.
- GigE 12-bit supports 15.29 fps.
- Required throughput at 10 fps:
  $$7.520256\times10=75.20\text{ MB/s}$$
- GigE provides 115 MB/s, which is sufficient.
- 12-bit provides approximately 61 grey levels for the crack contrast, compared with only 4 grey levels in 8-bit.

Conclusion:

The most economical configuration is a 12-bit camera using GigE Vision at 10 fps. It satisfies the production requirement, provides sufficient contrast resolution for detecting small cracks, and avoids the additional cost of USB3 Vision.

Question 2 (CO1, BTL3, 10 Marks) – Pinhole Projection Model for a Speed-Enforcement Camera

A traffic authority installs a roadside camera to estimate vehicle speed from image motion. The camera uses a 12 mm lens and a 1920 × 1080 sensor with square pixels of pitch 4.8 μm; the principal point lies at the image centre, and lens distortion has been corrected. Vehicles pass perpendicular to the optical axis in a lane whose distance from the camera centre is Z = 40 m. A reference point on a car has camera coordinates (X, Y, Z) = (1.2 m, −0.8 m, 40 m). Between two consecutive frames

captured at 30 fps, the car's licence plate moves 26 pixels horizontally in the image. The pixel localisation uncertainty is ±1 pixel.

(a) Determine the focal length in pixels, write the intrinsic matrix K, and project the reference point to pixel coordinates (u, v). [3 Marks]

(b) Compute the image length of a 4.5 m car at Z = 40 m and Z = 80 m, and the ground distance represented by one pixel at each depth. [3 Marks]

(c) Estimate the vehicle speed in m/s and km/h from the 26-pixel displacement. [2 Marks]

(d) Evaluate the speed error caused by the ±1 pixel uncertainty and by a 2 m error in assumed depth (42 m instead of 40 m), and justify how the measurement should be improved. [2 Marks]

ANSWER:

Pinhole Projection Model for a Speed-Enforcement Camera

(a) Focal Length, Intrinsic Matrix and Pixel Coordinates [3 Marks]

Given:

- Lens focal length = 12 mm
- Sensor resolution = 1920 × 1080 pixels
- Pixel pitch = 4.8 μm = 0.0048 mm
- 3D point = (X, Y, Z) = (1.2, −0.8, 40) m

1. Focal length in pixels

Formula:

$$ f_{px}=\frac{f}{\text{Pixel pitch}} $$

$$ =\frac{12}{0.0048}=\boxed{2500\text{ pixels}} $$

2. Intrinsic matrix K

Principal point:

$$ c_x=\frac{1920}{2}=960 $$

$$ c_y=\frac{1080}{2}=540 $$

Intrinsic matrix:

$$ K= \begin{bmatrix} 2500&0&960\\ 0&2500&540\\ 0&0&1 \end{bmatrix} $$

3. Project the 3D point into image coordinates

Formula:

$$ u=f_{px}\frac{X}{Z}+c_x $$

$$ v=f_{px}\frac{Y}{Z}+c_y $$

Substituting:

$$ u=2500\left(\frac{1.2}{40}\right)+960 $$

$$ \boxed{u=1035} $$

$$ v=2500\left(\frac{-0.8}{40}\right)+540 $$

$$ \boxed{v=490} $$

Final pixel coordinates: (1035, 490).

(b) Image Length and Ground Distance per Pixel [3 Marks]

Given:

- Car length = 4.5 m
- Focal length = 2500 pixels

Formula:

$$ \text{Image length}=\frac{f_{px}\times L}{Z} $$

1. At Z = 40 m

$$ =\frac{2500\times4.5}{40} $$

$$ \boxed{281.25\text{ pixels}} $$

Ground distance per pixel:

$$ =\frac{Z}{f_{px}}=\frac{40}{2500} $$

$$ \boxed{0.016\text{ m/pixel}} $$

2. At Z = 80 m

$$ =\frac{2500\times4.5}{80} $$

$$ \boxed{140.625\text{ pixels}} $$

Ground distance per pixel:

$$ =\frac{80}{2500} $$

$$ \boxed{0.032\text{ m/pixel}} $$

Result:

- At 40 m, the car appears larger (281.25 pixels).
- At 80 m, the car appears smaller (140.625 pixels).
- As the distance increases, the image size decreases.

(c) Vehicle Speed Estimation [2 Marks]

Given:

- Horizontal displacement = 26 pixels
- Frame rate = 30 fps
- Depth = 40 m
- Focal length = 2500 pixels

Time between frames:

$$ \Delta t=\frac{1}{30}=0.0333\text{ seconds} $$

Ground displacement:

$$ \Delta X=\frac{\Delta u\times Z}{f_{px}} $$

$$ =\frac{26\times40}{2500} $$

$$ =0.416\text{ m} $$

Vehicle speed:

$$ V=\frac{\text{Distance}}{\text{Time}} $$

$$ =\frac{0.416}{1/30} $$

$$ \boxed{V=12.48\text{ m/s}} $$

Convert into km/h:

$$ =12.48\times3.6 $$

$$ \boxed{V=44.93\text{ km/h}} $$

Result: The estimated vehicle speed is 44.93 km/h.

(d) Speed Error and Measurement Improvement [2 Marks]

1. Error due to ±1 pixel uncertainty

Pixel displacement = 26 ± 1 pixels.

Speed uncertainty:

$$ \Delta V=\frac{1\times40}{2500}\times30 $$

$$ =\boxed{\pm0.48\text{ m/s}} $$

In km/h:

$$ =0.48\times3.6 $$

$$ =\boxed{\pm1.728\text{ km/h}} $$

Therefore, estimated speed is:

$$ 44.93\pm1.73\text{ km/h} $$

2. Error due to 2 m depth error

Actual depth = 40 m

Assumed depth = 42 m

$$ V=\frac{26\times42}{2500}\times30 $$

$$ =13.104\text{ m/s} $$

Convert into km/h:

$$ =13.104\times3.6 $$

$$ =47.17\text{ km/h} $$

Speed error:

$$ =47.17-44.93 $$

$$ =\boxed{2.24\text{ km/h}} $$

Percentage error:

$$ =\frac{42-40}{40}\times100 $$

$$ =\boxed{5\%} $$

Methods to improve measurement

- Use accurate camera calibration to determine the actual vehicle depth.
- Use higher-resolution images and sub-pixel feature localisation to reduce pixel uncertainty.
- Capture multiple frames and calculate average displacement to improve speed accuracy.

Conclusion: Accurate depth measurement and proper camera calibration are important for obtaining reliable vehicle speed estimates.

Question 3 (CO1, BTL3, 10 Marks) – Illumination–Reflectance Model for Warehouse Barcode Reading

An automated warehouse reads barcodes on parcels moving under a skylight. The image formation follows $f(x, y) = i(x, y) \cdot r(x, y)$. The illumination is 900 lm/m² in the sunlit zone and 150 lm/m² in a shadowed zone created by an overhead rack. Reflectances are 0.85 for the white label, 0.05 for the black bars and 0.35 for the cardboard box. The camera maps f linearly to an 8-bit grey level using $g = \text{round}(255 \times f / 800)$; values of f above 800 saturate at 255. The current decoder uses a fixed global threshold of 128.

(a) Compute f(x, y) and the grey level g for the three surfaces in both zones and present the results in a table. [3 Marks]

(b) Compute the absolute grey-level difference and the Michelson contrast $(I_{max} - I_{min}) / (I_{max} + I_{min})$ between the white label and black bars in each zone. Interpret the result. [3 Marks]

(c) Apply the global threshold of 128 to both zones and determine whether the barcode can be decoded. Explain why no single global threshold can work. [2 Marks]

(d) Using $\ln f = \ln i + \ln r$, show that the label-to-bar ratio is independent of illumination, and justify a suitable processing strategy for the decoder. [2 Marks]

ANSWER:

Illumination–Reflectance Model for Warehouse Barcode Reading

(a) Calculate Image Intensity and Grey Level [3 Marks]

Given:

- Sunlit illumination = 900 lm/m²
- Shadow illumination = 150 lm/m²
- White label reflectance = 0.85
- Black bars reflectance = 0.05
- Cardboard reflectance **= 0.35**

Formula:

$$ f(x,y)=i(x,y)\times r(x,y) $$

$$ g=\operatorname{round}\left(\frac{255\times f}{800}\right) $$

1. Sunlit zone (900 lm/m²)

White label:

$$ f=900\times0.85=765 $$

$$ g=\operatorname{round}(255\times765/800)=244 $$

Black bars:

$$ f=900\times0.05=45 $$

$$ g=\operatorname{round}(255\times45/800)=14 $$

Cardboard box:

$$ f=900\times0.35=315 $$

$$ g=\operatorname{round}(255\times315/800)=100 $$

2. Shadowed zone (150 lm/m²)

White label:

$$ f=150\times0.85=127.5 $$

$$ g=\operatorname{round}(255\times127.5/800)=41 $$

Black bars:

$$ f=150\times0.05=7.5 $$

$$ g=\operatorname{round}(255\times7.5/800)=2 $$

Cardboard box:

$$ f=150\times0.35=52.5 $$

$$ g=\operatorname{round}(255\times52.5/800)=17 $$

Final Table:

Surface	Sunlit f	Sunlit g	Shadow f	Shadow g
White label	765	244	127.5	41
Black bars	45	14	7.5	2
Cardboard	315	100	52.5	17

(b) Absolute Grey-Level Difference and Michelson Contrast [3 Marks]

Formula:

Absolute grey-level difference:

$$ D=I_{max}-I_{min} $$

Michelson contrast:

$$ C=\frac{I_{max}-I_{min}}{I_{max}+I_{min}} $$

1. Sunlit zone

Grey levels:

- White label = 244
- Black bars = 14

Absolute difference:

$$D=244-14=\boxed{230}$$

Michelson contrast:

$$C=\frac{244-14}{244+14}$$

$$=\frac{230}{258}$$

$$\boxed{C=0.8915=89.15\%}$$

2. Shadowed zone

Grey levels:

- White label = 41
- Black bars = 2

Absolute difference:

$$D=41-2=\boxed{39}$$

Michelson contrast:

$$C=\frac{41-2}{41+2}$$

$$=\frac{39}{43}$$

$$\boxed{C=0.9070=90.70\%}$$

Interpretation:

1. The sunlit zone has a grey-level difference of 230.
2. The shadowed zone has a smaller difference of 39.
3. Both zones have high Michelson contrast, but the shadowed zone has lower absolute intensity, making it more sensitive to camera noise.

(c) Global Thresholding and Barcode Decoding [2 Marks]

Given:

Global threshold = 128

Rule:

- Grey level greater than 128 → White (1)
- Grey level less than or equal to 128 → Black (0)

1. Sunlit zone

- White label = 244 → White
- Black bars = 14 → Black
- Barcode can be decoded successfully.

2. Shadowed zone

- White label = 41 → Black

- Black bars = 2 → Black
- Both become black.

Therefore, the barcode cannot be decoded in the shadowed zone.

Why global threshold fails:

1. Illumination changes between sunlit and shadowed regions.
2. The same threshold cannot correctly separate white and black surfaces in both zones.
3. Hence, a fixed global threshold is unsuitable for varying illumination conditions**.**

(d) Illumination Independence and Processing Strategy [2 Marks]

Given:

$$f=i\times r$$

Taking logarithm on both sides:

$$\ln f=\ln i+\ln r$$

For white label:

$$\ln f_w=\ln i+\ln r_w$$

For black bars:

$$\ln f_b=\ln i+\ln r_b$$

Subtracting:

$$\ln f_w-\ln f_b =\ln r_w-\ln r_b$$

Therefore,

$$\boxed{\ln\left(\frac{f_w}{f_b}\right) =\ln\left(\frac{r_w}{r_b}\right)}$$

The illumination term is cancelled. Thus, the label-to-bar intensity ratio is independent of illumination.

Suitable processing strategy:

- Apply illumination normalization to reduce brightness variations.
- Use adaptive thresholding instead of a fixed global threshold.
- Apply local contrast enhancement to improve barcode detection in shadowed regions.

Conclusion: Adaptive thresholding and illumination normalization help the barcode decoder work accurately in both sunlit and shadowed areas.

Question 4 (CO1, BTL3, 10 Marks) – Pixel Connectivity Analysis for PCB Track Continuity

An automated optical inspection system checks whether copper tracks on a printed circuit board are electrically continuous. After binarisation, a 6 × 6 region gives the image below, where 1 is copper and 0 is substrate (V = {1}). The test points are p = (R1, C1) and q = (R6, C6), and an intermediate

pad is s = (R4, C4). The design rule states that diagonal-only contacts are not reliable conductors, so the engineer must decide which adjacency definition models the board correctly

Row / Col	C1	C2	C3	C4	C5	C6
R1	1	1	0	0	0	0
R2	0	1	0	0	1	1
R3	0	0	1	0	1	0
R4	0	0	1	1	0	0
R5	1	0	0	1	0	1
R6	1	1	0	0	1	1

(a) Compute the Euclidean, city-block (D_4) and chessboard (D_8) distances between p and q. [3 Marks]
(b) Determine the number of connected components of copper under 4-adjacency and under 8-adjacency, and list the pixels of each component under 8-adjacency. [3 Marks]
(c) Find the length of the shortest 8-path and the shortest m-path from p to q, and from p to s. [2 Marks]
(d) Under 4-adjacency, p and q are not connected. Using your results, justify which adjacency correctly models electrical continuity and state the inspection verdict. [2 Marks]

ANSWER:

Pixel Connectivity Analysis for PCB Track Continuity
Given:

- Image size = 6 × 6
- V = 1 (Copper pixels)
- Starting point p = (R1, C1) = (1,1)
- Ending point q = (R6, C6) = (6,6)
- Intermediate pad s = (R4, C4) = (4,4)

(a) Euclidean, City-Block and Chessboard Distances
1. Euclidean Distance

Formula:

$$D_E=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$=\sqrt{(6-1)^2+(6-1)^2}$$

$$=\sqrt{25+25}$$

$$\boxed{D_E=7.07\text{ pixels}}$$

2. City-Block Distance (D4)

Formula:

$$D_4=|x_2-x_1|+|y_2-y_1|$$

$$ =|6-1|+|6-1| $$

$$ =5+5 $$

$$ \boxed{D_4=10\text{ pixels}} $$

3. Chessboard Distance (D8)

Formula:

$$ D_8=\max(|x_2-x_1|,|y_2-y_1|) $$

$$ =\max(5,5) $$

$$ \boxed{D_8=5\text{ pixels}} $$

Final Results:

Distance	Answer
Euclidean	7.07 pixels
City-Block (D4)	10 pixels
Chessboard (D8)	5 pixels

(b) Connected Components under 4 and 8 Adjacency

Under 4-adjacency

4-adjacency considers only horizontal and vertical neighbouring pixels.

The copper pixels form 5 connected components.

- Component 1: (1,1), (1,2), (2,2)
- Component 2: (3,3), (4,3), (4,4), (5,4)
- Component 3: (2,5), (2,6), (3,5)
- Component 4: (5,6), (6,5), (6,6)
- Component 5: (5,1), (6,1), (6,2)

Therefore,

$$ \boxed{\text{4-connected components}=5} $$

2. Under 8-adjacency

8-adjacency considers horizontal, vertical and diagonal neighbouring pixels.

The copper pixels form 2 connected components.

- Component 1: (1,1), (1,2), (2,2), (3,3), (4,3), (4,4), (5,4), (6,5), (6,6), (5,6), (2,5), (2,6), (3,5)
- Component 2: (5,1), (6,1), (6,2)

Therefore,

$$ \boxed{\text{8-connected components}=2} $$

Result: 8-adjacency produces fewer connected components because it also considers diagonal connections.

(c) Shortest 8-Path and m-Path

1. Shortest 8-path from p to q

One shortest path is:

(1,1) → (2,2) → (3,3) → (4,3) → (5,4) → (6,5) → (6,6)

Number of steps = 6.

$$\boxed{\text{Shortest 8-path}=6}$$

2. Shortest m-path from p to q

One shortest m-path is:

(1,1) → (1,2) → (2,2) → (3,3) → (4,3) → (4,4) → (5,4) → (6,5) → (6,6)

Number of steps = 8.

$$\boxed{\text{Shortest m-path}=8}$$

3. Shortest 8-path from p to s

Path:

(1,1) → (2,2) → (3,3) → (4,4)

Number of steps = 3.

$$\boxed{\text{Shortest 8-path}=3}$$

4. Shortest m-path from p to s

Path:

(1,1) → (1,2) → (2,2) → (3,3) → (4,3) → (4,4)

Number of steps = 5.

$$\boxed{\text{Shortest m-path}=5}$$

Result: m-adjacency avoids ambiguous diagonal connections when common 4-neighbours are also copper pixels.

(d) Adjacency Selection and Inspection Verdict

- 4-adjacency considers only horizontal and vertical connections. Under this method, p and q are not connected.
- 8-adjacency considers diagonal connections also. It shows a path between p and q, but diagonal-only contacts are not reliable electrical conductors.
- Therefore, 4-adjacency is the correct model for checking electrical continuity according to the given design rule.

Final Inspection Verdict:

FAIL – Discontinuous Track

The PCB track is considered electrically discontinuous because p and q are not connected under 4-adjacency.

Conclusion: 4-adjacency is suitable for PCB electrical continuity inspection because diagonal-only contacts should not be treated as reliable copper connections.

Question 5 (CO1, BTL3, 10 Marks) – Levels of Processing in a Number-Plate Recognition System

A smart-parking operator deploys an automatic number-plate recognition (ANPR) system on an embedded processor. Each 1280 × 720 colour frame (24 bits per pixel) passes through five stages with measured execution times: noise filtering and contrast correction (6 ms), plate localisation by segmentation (14 ms), character segmentation (5 ms), character recognition using a classifier (9 ms) and database matching for access decision (4 ms). Plate localisation outputs a 240 × 60 greyscale ROI (8 bits per pixel), character segmentation outputs eight binary character images of 20 × 40 pixels each, and the final output is a 10-character ASCII string. The camera delivers 25 fps.

(a) Classify each stage as low-, mid- or high-level processing and compute the end-to-end latency and the maximum throughput for sequential and for fully pipelined execution. [3 Marks]

(b) Compute the data volume at the input and after each level, and the overall data reduction ratio. [3 Marks]

(c) The operator upgrades to a 1920 × 1080 camera. Assuming the first two stages scale with pixel count and the others are unchanged, determine whether 25 fps is achievable sequentially and pipelined. [2 Marks]

(d) Justify an architecture for the upgraded system and explain, using this pipeline, where image processing ends and computer vision begins. [2 Marks]

ANSWER:

(a) Classification, latency and throughput

Classification of five stages:

Processing stage	Level
Noise filtering and contrast correction	Low-level
Plate localisation by segmentation	Mid-level
Character segmentation	Mid-level
Character recognition using classifier	High-level
Database matching for access decision	High-level

1. End-to-end latency (Sequential)

Total execution time:

$$T=6+14+5+9+4$$

$$\boxed{T=38\ ms}$$

2. Maximum throughput (Sequential)

$$ Throughput=\frac{1000}{38} $$

$$ \boxed{26.32\ fps} $$

3. Maximum throughput (Fully pipelined)

The slowest stage determines the pipeline throughput.

$$ T_{max}=\max(6,14,5,9,4)=14\ ms $$

$$ Throughput=\frac{1000}{14} $$

$$ \boxed{71.43\ fps} $$

Therefore, the pipelined system provides higher throughput.

(b) Data volume and data reduction ratio [3 Marks]

1. Input image data

$$ 1280\times720\times24 =22,118,400\ bits $$

$$ =2,764,800\ bytes $$

2. After noise filtering and contrast correction

Image size remains unchanged.

$$ 1280\times720\times24 =22,118,400\ bits $$

3. After plate localisation

ROI = 240 × 60, greyscale = 8 bits.

$$ 240\times60\times8=115,200\ bits $$

$$ =14,400\ bytes $$

4. After character segmentation

Eight binary images of 20 × 40 pixels.

$$ 8\times20\times40\times1 =6,400\ bits $$

$$ =800\ bytes $$

5. Final output

10-character ASCII string (8 bits per character).

$$ 10\times8=80\ bits $$

$$ =10\ bytes $$

Data volume summary:

Stage	Data volume
Input colour image	2,764,800 bytes
After noise filtering	2,764,800 bytes
After plate localisation	14,400 bytes
After character segmentation	800 bytes
Final ASCII output	10 bytes

Overall data reduction ratio

$$= \frac{Input\ data}{Final\ output}$$

$$= \frac{2,764,800}{10}$$

$$\boxed{276,480:1}$$

(c) Upgraded camera: 1920 × 1080 [2 Marks]

Original resolution = 1280 × 720.

New resolution = 1920 × 1080.

Scaling factor:

$$\frac{1920\times1080}{1280\times720}=2.25$$

The first two stages scale with pixel count.

- Noise filtering: $6\times2.25=13.5\ ms$
- Plate localisation: $14\times2.25=31.5\ ms$

Other stages remain unchanged.

1. Sequential execution

$$T=13.5+31.5+5+9+4$$

$$\boxed{T=63\ ms}$$

$$Throughput=\frac{1000}{63}=15.87\ fps$$

25 fps not achievable

2. Fully pipelined execution

The slowest stage is 31.5 ms.

$$Throughput=\frac{1000}{31.5}$$

$$\boxed{31.75\ fps}$$

25 fps achievable

Therefore, pipelined execution can achieve the required 25 fps.

(d) Architecture justification and Image Processing vs Computer Vision

Recommended architecture:

A fully pipelined architecture is suitable for the upgraded ANPR system because:

- Multiple frames can be processed simultaneously at different stages.
- It provides 31.75 fps, which is greater than the required 25 fps.
- It reduces processing delays and improves system efficiency.
- It is suitable for real-time number-plate recognition.

Image Processing vs Computer Vision:

- Image Processing: Includes noise filtering, contrast correction, plate localisation and character segmentation. It improves the image and extracts useful regions.

- Computer Vision: Includes character recognition and database matching. It identifies the number plate and makes an access decision.

  Conclusion: Image processing prepares and extracts image information, whereas computer vision interprets the information and makes meaningful decisions.